# Hardware Introduction

This document provides a detailed introduction to the hardware components, functions, and usage methods of the LAFVIN ESP32S3 AIChatBot.

## Main Controller - ESP32-S3

ESP32-S3 is the core processor of this project, providing powerful computing capabilities and rich interfaces.



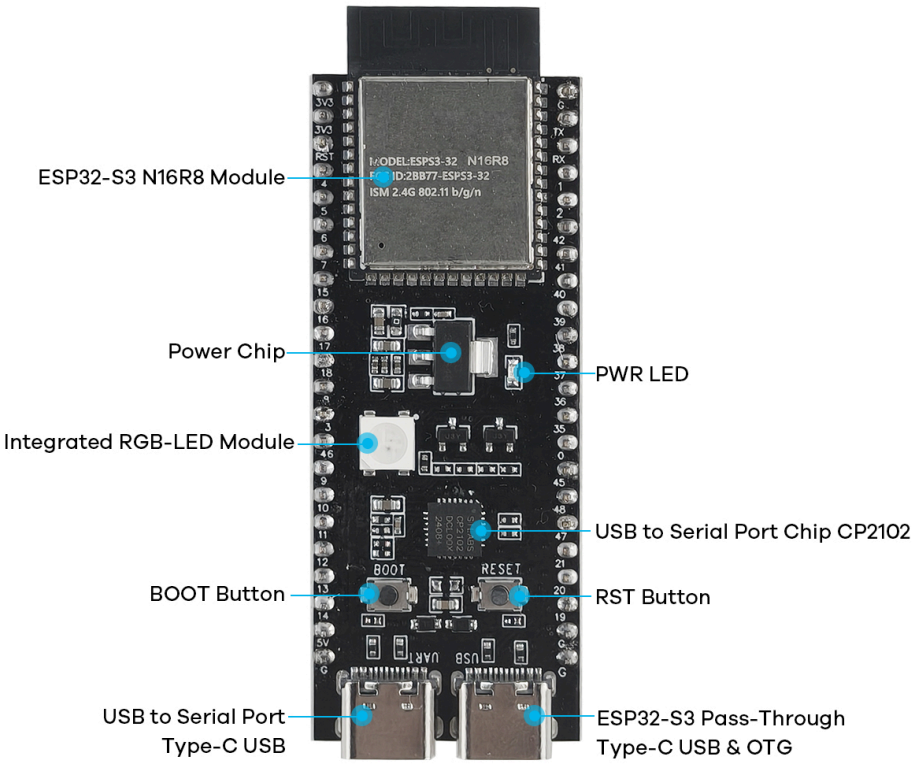

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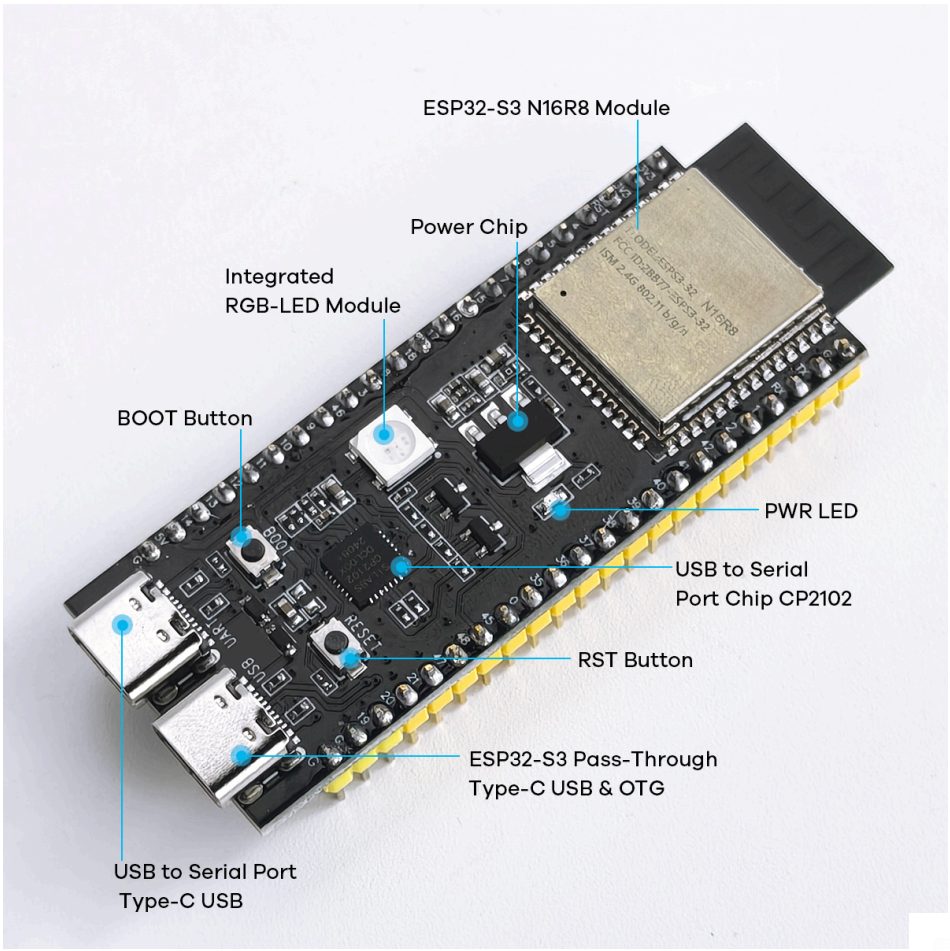
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- Integrated Wi-Fi and Bluetooth dual-mode communication
- Rich peripheral interfaces, supporting various communication protocols
- Low power design, suitable for battery-powered applications
- Built-in neural network accelerator, supporting local AI computation

### Technical Specifications

| | |
|---|---|
| Processor | Dual-core Xtensa LX7, frequency up to 240MHz |
| Memory | 512KB SRAM, 8MB PSRAM |
| Storage | Supports up to 16MB Flash |
| Wi-Fi | IEEE 802.11 b/g/n, supports the 2.4 GHz band |
| Bluetooth | Bluetooth 5.0 and BLE |
| GPIO | Up to 45 programmable GPIOs |
| Interfaces | SPI, I2C, I2S, UART, USB OTG |
| Operating Voltage | 3.0-3.6V |
| Operating Temperature | -40°C to +85°C |

## Display Module

This project uses a 2.0-inch 240*320 SPI LCD display, based on the ST7789 controller, communicating with the ESP32-S3 via SPI interface.




### Key Features

- Resolution: 240*320 pixels
- RGB color display

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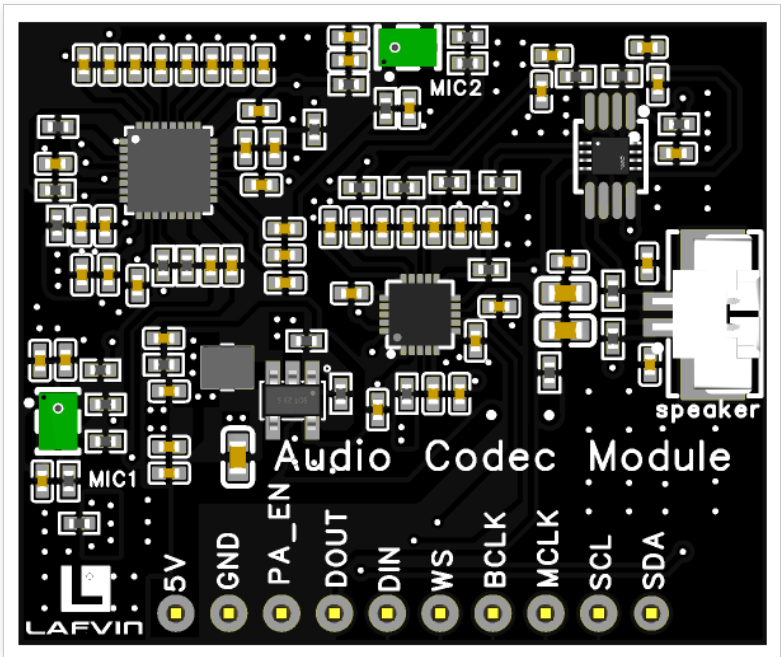
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The display is mainly used for showing system status, message content, and interaction prompts.

## LAFVIN Audio Codec Module

The LAFVIN ESP32S3 AIChatBot utilizes the LAFVIN Audio Codec Module, which supports voice input and output functions, enabling the device to perform voice interactions.




### Key Features

- High Signal-to-Noise Ratio: Ensures clear and accurate sound pickup.
- Built-in Digital Signal Processing (DSP): Enhances audio quality and performance.
- I2S Interface: Directly outputs digital audio signals, compatible with microcontrollers and processors.
- Dual Omnidirectional Microphones: Wide coverage for robust voice capture.
- Sufficient Volume Output: Ensures clear audio playback even in noisy environments.

## Buttons and Interaction

The device provides physical buttons for simple operations and control:

1. The button connected to GPIO20 is for scrolling text upward
2. The button connected to GPIO19 is for scrolling text downward

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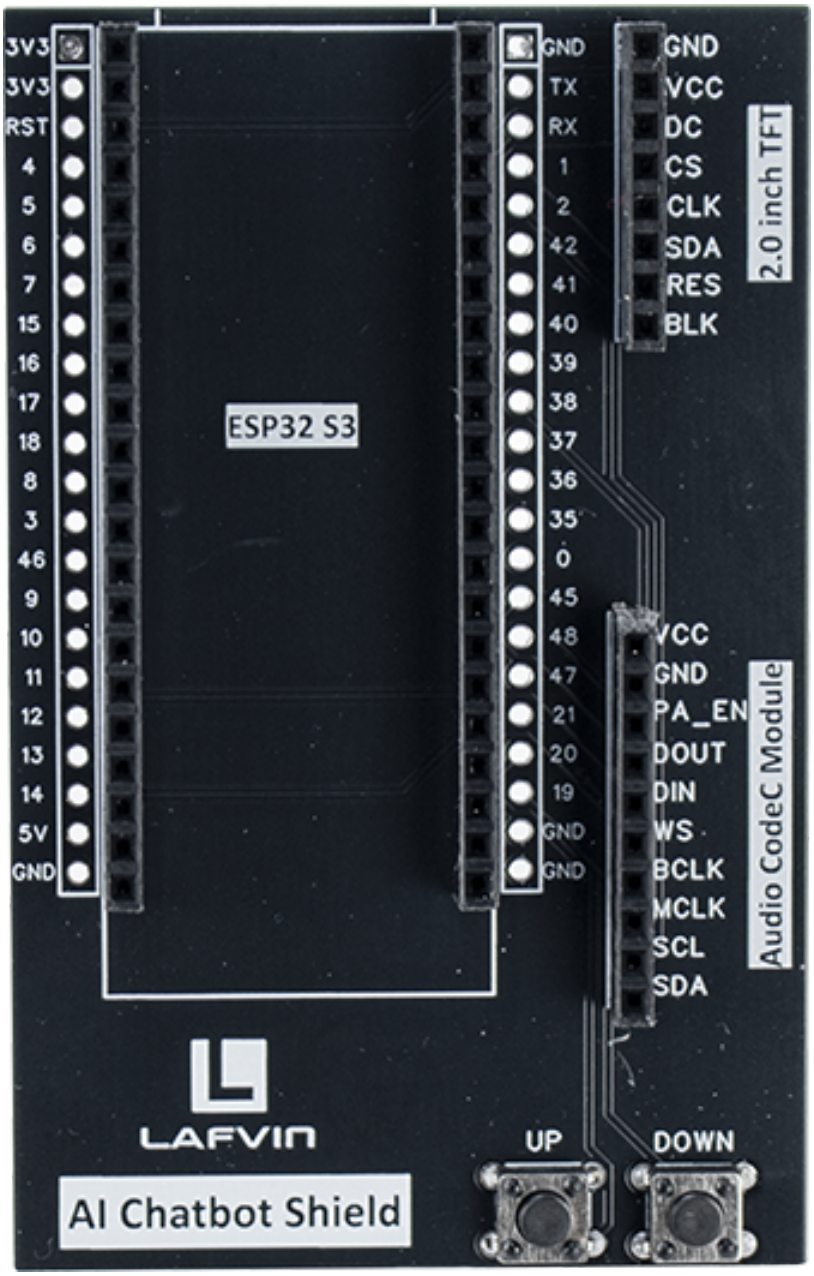
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# AI Chatbot Shield

The AI Chatbot Shield is an expansion board for the LAFVIN ESP32S3 AIChatBot, designed to simplify the wiring process and provide a plug-and-play solution for implementing AI Chatbot functionality.




# Hardware Connection Diagram

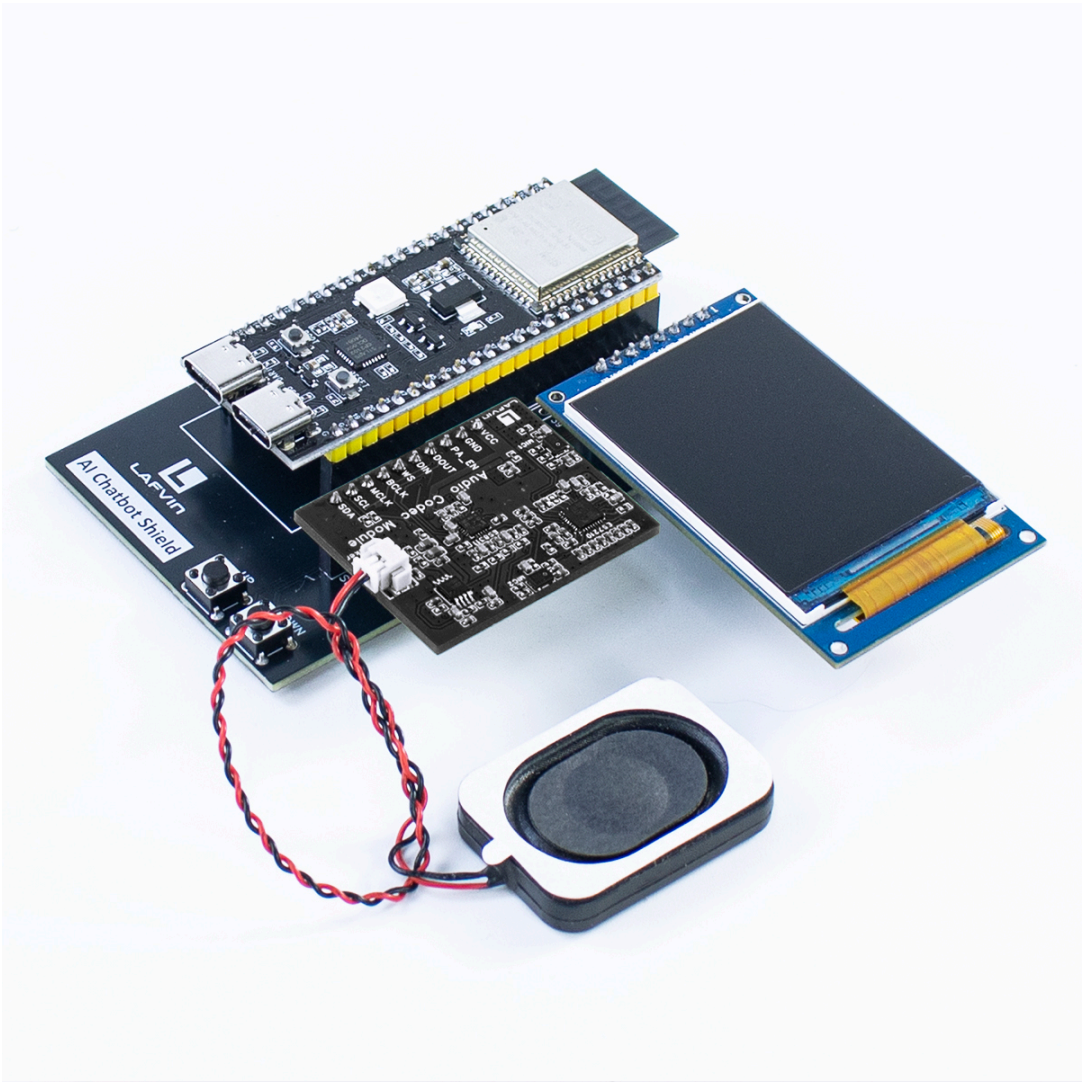
The following diagram shows how each hardware module is connected:

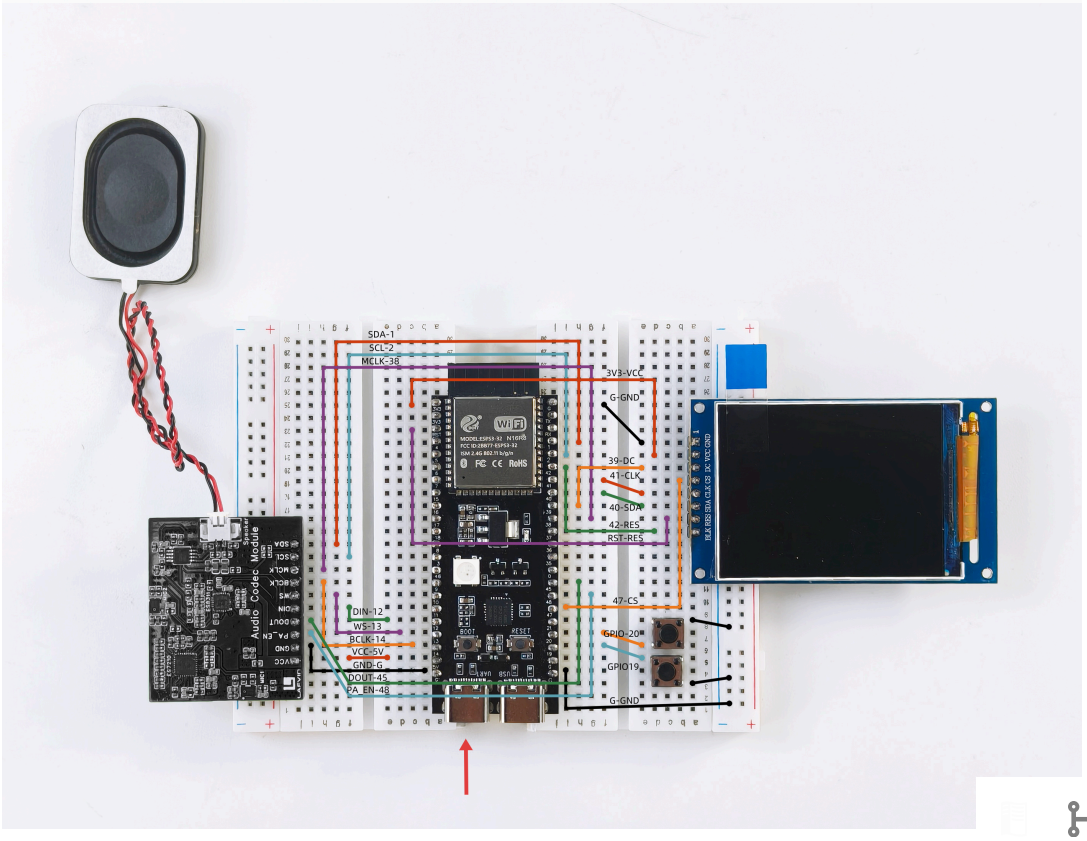
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